20. $E_{cell}=E^{o}_{cell}-\frac{0.0591}{2}\log\frac{[zn^{2+}]}{[Cu^{2+}]}$

$E_1=E^{o}_{cell}-\frac{0.0591}{2}\log\frac{10^{-2}}{1}$

$E_1=E^{o}_{cell}+0.0591.....(1)$

$Zn(s) \rightarrow Zn^{2+}(aq)+2e^{-}$

$Cu^{2+}(aq)+2e^{-} \rightarrow Cu(s)$

$Zn(s)+Cu^{2+}(aq) \rightarrow Zn^{2+}(aq)+Cu(s)$

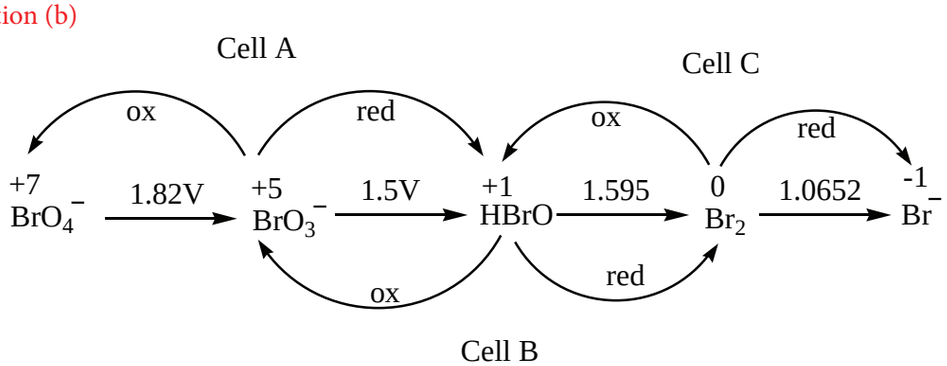
$E_2=E^{o}_{cell}-\frac{0.0591}{2}.\log\frac{1}{10^{-2}}$

$E_2=E^{o}_{cell}-0.0591.....(2)$

$\therefore E_1>E_2$

Option (b)

21.



$(E_{cell})_A=-1.82+1.5=-0.32V$

$(E_{cell})_B=-1.5+1.595=+0.095V$

$(E_{cell})_c=-1.595+1.0652=-0.529V$

$\therefore$ The species undergoing disproportionation is $HBrO$ (Option D)

Short answer

8. Given

$C = 0.01M$ $\quad \lambda^{o}_{cation} = 248.2\ S\ cm^2\ mol^{-1}$

$K= 1.5\times10^{-4}\ S\ cm^{-1}$ $\quad \lambda^{o}_{anion} = 51.8\ S\ cm^2\ mol^{-1}.$

1. Molar conductivity

$\Lambda^{\circ}_{m} = \frac{K(sm^{-1})\times10^{-3}}{C\ (in\ M)}\ mol^{-1}\ m^3$ $\quad K = 1.5\times10^{-4}S\ cm^{-1}$

$= \frac{1.5\times10^{2}\times10^{-3}}{0.01}\ S\ mol^{-1}\ m^2$ $\quad 1\ cm^{-1} = 10^{2}m^{-1}$

$= 1.5\times10^{-3}\ S\ m^2\ mol^{-1}$ $\quad = 1.5\times10^{2}$

2. Degree of dissociation $\alpha = \frac{\Lambda^{\circ}}{\Lambda^{\circ}_{\infty}}$

$\Lambda^{o}_{\infty}=\lambda^{o}_{cation}+\lambda^{o}_{anion}$

$= (248.2+51.8)\ S\ cm^2\ mol^{-1}$

$= 300\ S\ cm^2\ mol^{-1}$

$= 300\times10^{-14}\ s\ m^2\ mol^{-1}$

Answers.indd 308 11-01-2021 17:36:37